ML PROJECT

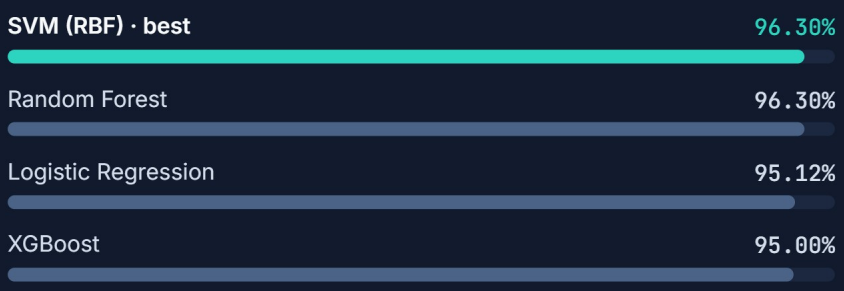
569 samples · 30 features

Breast Cancer Detection with ML

I trained and compared 4 models to classify tumors as malignant or benign.

97.4%	100%	0.995
Test accuracy	Precision	ROC-AUC

Model comparison — F1-score, test set (114)



Stack: `Python` `Scikit-learn` `XGBoost` `Pandas` `5-fold CV`

Muhammad Awais
AI Engineer · Machine Learning

Code on GitHub →

HOW IT WORKS

From raw data to diagnosis

1. **Load the data**
   569 tumors · 212 malignant · 357 benign
2. **Split & scale**
   Stratified 80 / 20 split · StandardScaler
3. **Train 4 models**
   Logistic Regression · Random Forest · SVM · XGBoost
4. **Cross-validate**
   5-fold stratified CV on the training set
5. **Evaluate on unseen data**
   Accuracy · Precision · Recall · F1 · ROC-AUC

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BEST MODEL · SVM

111 of 114 correct

Confusion matrix on the unseen test set.

	Predicted malignant	Predicted benign
Actually malignant	39 Caught	3 Missed
Actually benign	0 False alarms	72 Correctly cleared

In healthcare, the 3 missed cases matter most. Next: tune the threshold to push recall above 92.9%.

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WHAT THE MODEL LEARNED

Size & shape tell the story

Top 5 features by Random Forest importance.



Larger, more irregular cells were the strongest signal of malignancy, which matches what pathologists look for.

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